UNIT 1 MINI PROJECT

Enterprise Student Record and Academic Management Platform

Design Thinking and Product Planning — Consolidated Deliverables

Prepared as part of the Design Thinking and Product Planning phase, preceding Agile development in subsequent units.

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Project Overview

Objective: Apply the Design Thinking methodology to design a software solution for managing student records and academic information within a higher education institution.

Rather than beginning with coding, this unit focuses on understanding the educational domain, identifying stakeholders, analyzing user needs, defining software requirements, and preparing the product vision and roadmap. The output of this unit is a well-planned software product ready for Agile planning and development in subsequent units.

Scope of this Document

This document consolidates all sixteen deliverables required for Unit 1:

- Business Requirements Document (BRD)
- Product Vision Document
- Stakeholder Analysis Report
- User Persona Report
- Functional Requirements Specification (FRS)
- Non-Functional Requirements Specification (NFRS)
- Design Thinking Report
- IBM Design Thinking Report
- IBM Loop Documentation
- MVP Definition
- Product Roadmap
- Initial System Architecture Diagram
- Project Charter
- Project Repository Initialization
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Deliverable 1

1. Business Requirements Document (BRD)

1.1 Purpose

This document defines the business needs and requirements for the Enterprise Student Record and Academic Management Platform (“the Platform”), a software system intended to digitize and streamline academic administration for a higher education institution.

1.2 Background and Problem Statement

Higher education institutions currently manage student admissions, course registration, attendance, examinations, and results through manual, paper-based, or fragmented spreadsheet-based processes. This leads to duplicated data entry, delayed report generation, inconsistent attendance records, difficulty tracking academic performance over time, and a poor experience for students, faculty, and administrators alike.

Manual record management specifically causes the following problems:

- Data duplication and inconsistency across departments (admissions, examination cell, faculty registers).
- Delays in generating mark sheets, transcripts, and attendance reports.
- Difficulty tracking a student's academic history across semesters.
- No centralized, real-time visibility for administrators into institution-wide academic metrics.
- High dependency on physical records, increasing risk of loss or damage.

1.3 Business Objectives

- Centralize student academic data into a single, authoritative system.
- Reduce administrative turnaround time for registration, attendance, and result processing.
- Improve data accuracy and auditability of academic records.
- Provide role-based self-service access for students, faculty, and administrators.
- Enable data-driven decision-making through consolidated reporting.

1.4 Scope

In Scope

- Student registration and profile management.
- Course registration and enrollment.
- Attendance recording and tracking.
- Examination scheduling and result management.
- Report generation (attendance, academic performance, transcripts).

Out of Scope (for this phase)

- Fee payment and financial accounting integration (future phase).
- Hostel and transport management.
- Alumni relationship management.

1.5 Business Requirements

ID	Requirement	Business Justification
BR-01	The system shall maintain a single digital record per student.	Eliminates duplication across departments.
BR-02	The system shall support online course registration.	Reduces manual registration workload.
BR-03	The system shall record and track daily attendance.	Enables timely intervention for low attendance.
BR-04	The system shall manage examination scheduling and result publishing.	Speeds up result turnaround.
BR-05	The system shall generate academic and administrative reports on demand.	Supports institutional decision-making.
BR-06	The system shall provide role-based access control.	Protects sensitive academic data.

1.6 Assumptions and Constraints

- The institution has stable internet connectivity across campus.
- Existing student data (if any) will be migrated from spreadsheets/legacy systems.
- The system will be delivered incrementally across the phases defined in the Product Roadmap.

Deliverable 2

2. Product Vision Document

2.1 Vision Statement

To become the single, trusted digital backbone for academic administration — empowering students, faculty, and administrators with accurate, real-time academic information and eliminating manual record-keeping in higher education institutions.

2.2 Mission Statement

To design and deliver a secure, scalable, and easy-to-use platform that simplifies student registration, attendance, examinations, and reporting, so that institutions can focus on educational outcomes rather than administrative overhead.

2.3 Product Positioning Statement

Element	Description
For	Higher education institutions and their students, faculty, and administrative staff
Who	Need a reliable, centralized way to manage academic records
The	Enterprise Student Record and Academic Management Platform
Is a	Web-based academic management system
That	Centralizes registration, attendance, examinations, and reporting in one platform
Unlike	Manual, paper-based, or disconnected spreadsheet processes
Our product	Provides real-time, role-based, auditable access to academic data for every stakeholder

2.4 Business Objectives

- Reduce administrative processing time for academic records by a measurable margin each semester.
- Achieve high user adoption among students and faculty within the first two semesters of rollout.
- Establish a single source of truth for academic data across the institution.

2.5 Success Criteria

- Faster turnaround for registration, attendance capture, and result publication compared to the manual process.
- Reduction in data discrepancies and record-correction requests.
- Positive usability feedback from students, faculty, and administrators.
- Successful completion of the MVP release, followed by adoption of subsequent roadmap phases.

Deliverable 3

3. Stakeholder Analysis Report

The table below identifies the primary stakeholders of the Platform, their interest in the system, their level of influence over decisions, and their core expectations.

Stakeholder	Role / Interest	Influence	Expectations
Students	Primary end users — register for courses, view attendance, results	Medium	Easy-to-use self-service portal; timely, accurate information
Faculty Members	Record attendance, enter marks, view class rosters	Medium	Fast data entry; minimal administrative overhead
Academic Administrators	Oversee admissions, registration, and academic policy	High	Institution-wide visibility and control; accurate reporting
Examination Cell Staff	Schedule exams, process and publish results	High	Reliable, auditable result-processing workflow
IT Support Personnel	Maintain, deploy, and support the system	Medium	Maintainable, well-documented, secure system
Institution Management	Sponsors the project; reviews outcomes	High	Measurable improvement in efficiency and data quality

3.1 Stakeholder Engagement Approach

- Students and Faculty: consulted through interviews/surveys to shape usability requirements.
- Academic Administrators and Examination Cell: consulted for process and compliance requirements.
- IT Support: consulted for technical constraints, security, and maintainability requirements.

Deliverable 4

4. User Persona Report

4.1 Persona: The Student

Name / Role: Ananya Rao, Second-Year Undergraduate Student

Goals: Register for courses quickly, check attendance and results without visiting the office, receive timely notifications about exams.

Responsibilities: Maintain minimum attendance, complete course registration each semester, track academic progress.

Challenges: Long queues for manual registration, delayed access to attendance/result information, no single place to view academic history.

Expectations from the Platform: A simple mobile-friendly portal to register for courses, view attendance percentage, and download results/transcripts.

4.2 Persona: The Faculty Member

Name / Role: Dr. Vikram Sen, Assistant Professor

Goals: Mark attendance quickly during class, enter and submit marks efficiently, view class performance trends.

Responsibilities: Conduct classes, record attendance, evaluate and submit examination marks.

Challenges: Manual attendance registers are time-consuming; mark entry via spreadsheets is error-prone and hard to consolidate.

Expectations from the Platform: A fast attendance-marking interface, a structured mark-entry workflow with validation, and class-level reports.

4.3 Persona: The Academic Administrator

Name / Role: Meera Joshi, Academic Administrator

Goals: Maintain accurate institution-wide student records, generate reports for accreditation and management review, ensure policy compliance.

Responsibilities: Oversee admissions and registration data, coordinate between departments, ensure examination and result-publishing timelines are met.

Challenges: Consolidating data from multiple departments and formats; lack of a single dashboard for institution-wide metrics.

Expectations from the Platform: A centralized admin console with role-based access, configurable reports, and audit trails.

Deliverable 5

5. Functional Requirements Specification (FRS)

The functional requirements below describe what the system shall do, grouped by module.

5.1 Student Registration Module

ID	Requirement	Priority
FR-01	The system shall allow new students to submit registration details online.	High
FR-02	The system shall validate mandatory fields (name, contact, prior academic records) at registration.	High
FR-03	The system shall assign a unique student ID upon successful registration.	High
FR-04	The system shall allow students to enroll in courses offered for their program/semester.	High

5.2 Attendance Management Module

ID	Requirement	Priority
FR-05	The system shall allow faculty to mark attendance per class session.	High
FR-06	The system shall calculate attendance percentage per student per course.	High
FR-07	The system shall flag students falling below the minimum attendance threshold.	Medium
FR-08	The system shall allow students to view their own attendance record.	Medium

5.3 Examination and Result Management Module

ID	Requirement	Priority
FR-09	The system shall allow the examination cell to schedule examinations per course.	High
FR-10	The system shall allow faculty to enter and submit marks for evaluation.	High
FR-11	The system shall compute grades based on configurable grading rules.	Medium
FR-12	The system shall publish results to students upon administrative approval.	High

5.4 Reporting Module

ID	Requirement	Priority
FR-13	The system shall generate attendance reports by student, course, or class.	Medium

ID	Requirement	Priority
FR-14	The system shall generate academic performance reports and transcripts.	High
FR-15	The system shall allow administrators to export reports (PDF/Excel).	Medium

5.5 User and Access Management Module

ID	Requirement	Priority
FR-16	The system shall support role-based login for students, faculty, and administrators.	High
FR-17	The system shall allow administrators to create, update, and deactivate user accounts.	Medium

Deliverable 6

6. Non-Functional Requirements Specification (NFRS)

Category	Requirement
Security	All user data shall be protected via role-based access control, encrypted storage of sensitive fields, and secure authentication (e.g., hashed passwords, session management).
Usability	The interface shall be intuitive enough for a first-time student or faculty user to complete core tasks (registration, attendance marking) without formal training.
Reliability	The system shall be available during academic hours with graceful handling of errors and no data loss during normal operation.
Scalability	The system shall support growth in student and course volume across multiple academic years without redesign.
Performance	Core operations (login, attendance marking, report generation) shall complete within a few seconds under normal load.
Availability	The system shall target high uptime during the academic term, with maintenance windows scheduled outside peak hours.
Maintainability	The system shall be built using modular components to allow independent updates to individual modules (e.g., examinations, attendance).
Compliance	The system shall align with institutional data-retention and privacy policies for student records.
Auditability	The system shall maintain an audit trail for critical actions such as result publishing and record modification.

Deliverable 7

7. Design Thinking Report

The Design Thinking methodology was applied across its five stages to arrive at the Platform's design direction.

7.1 Empathize

Engaged with representative students, faculty, and administrators (through interviews and process observation) to understand pain points in the current manual academic-management process, such as long registration queues, delayed results, and scattered attendance records.

7.2 Define

Problem Statement: Students, faculty, and administrators at the institution lack a centralized, reliable system to manage academic records, leading to delays, data inconsistency, and administrative overhead across registration, attendance, and examination processes.

7.3 Ideate

Candidate solution ideas generated and evaluated:

- A centralized web platform covering registration, attendance, and results (selected direction).
- A set of independent departmental tools integrated later (rejected — perpetuates data fragmentation).
- A mobile-only application (partially adopted — responsive design instead of mobile-only, to support administrative desktop workflows).

7.4 Prototype

Low-fidelity wireframes and the initial system architecture (see Deliverable 12) were used to represent the registration flow, attendance-marking flow, and result-publishing flow, enabling early stakeholder feedback before development begins.

7.5 Test

Concepts and wireframes were reviewed with sample students, faculty, and administrators; feedback (e.g., preference for a simplified attendance-marking screen, and the need for downloadable transcripts) was folded into the Functional Requirements Specification and MVP Definition.

Deliverable 8

8. IBM Design Thinking Report

IBM Design Thinking principles were applied alongside the classic Design Thinking stages to keep the process grounded in outcomes and diverse perspectives.

8.1 Focus on User Outcomes

Requirements were framed around the outcomes each stakeholder needs — e.g., a student's outcome is “know my attendance and results without delay,” rather than simply “have a portal.” This outcome-first framing shaped the Functional Requirements Specification and MVP scope.

8.2 Restless Reinvention

The product roadmap treats the MVP as a starting point, not an end state: attendance thresholds, grading rules, and reporting formats are planned as configurable so the platform can keep evolving with institutional needs across roadmap phases.

8.3 Diverse Empowered Teams

Requirements gathering intentionally included perspectives from students, faculty, examination cell staff, academic administrators, and IT support, ensuring the design reflects both end-user needs and operational/technical constraints rather than a single viewpoint.

Deliverable 9

9. IBM Loop Documentation — Observe → Reflect → Make

Iteration 1 — Registration & Records

Observe: Students and administrators spend significant time on manual registration forms and duplicate data entry across departments.

Reflect: A single digital student record, created once at registration and reused across modules, would remove duplication and speed up onboarding.

Make: Defined the Student Registration functional requirements (FR-01–FR-04) and the single-record principle in the BRD.

Iteration 2 — Attendance

Observe: Paper attendance registers are slow to fill and make it hard to identify at-risk students in time.

Reflect: Digitizing attendance capture and automatically calculating percentages enables early, automated flagging of low attendance.

Make: Defined Attendance Management requirements (FR-05–FR-08), including automatic threshold flagging.

Iteration 3 — Examinations & Results

Observe: Result compilation and publishing is delayed due to manual mark aggregation across faculty and courses.

Reflect: A structured mark-entry and approval workflow shortens the path from evaluation to publishing.

Make: Defined Examination and Result Management requirements (FR-09–FR-12) and included result publishing as an MVP capability.

Deliverable 10

10. Minimum Viable Product (MVP) Definition

10.1 MVP Goal

Deliver the smallest usable version of the Platform that replaces the most painful manual processes — registration, attendance, and academic reporting — while establishing the centralized data model the rest of the roadmap builds on.

10.2 MVP Scope

- Student registration and profile management (FR-01–FR-04).
- Course enrollment for the current academic term.
- Attendance recording by faculty and attendance viewing by students (FR-05, FR-06, FR-08).
- Basic academic report generation (attendance summaries, student profile reports) (FR-13, FR-14 partial).
- Role-based login for students, faculty, and administrators (FR-16).

10.3 Explicitly Out of MVP Scope

- Full examination scheduling and result publishing workflow (planned for Phase 2).
- Advanced analytics and configurable report exports (planned for Phase 3).
- External integrations such as institution ERP or payment gateways (planned for later phases).

10.4 MVP Success Criteria

- A student can register, enroll in courses, and view their attendance without visiting an office.
- A faculty member can mark attendance for a class session in a few minutes or less.
- An administrator can generate a basic attendance/registration report on demand.

Deliverable 11

11. Product Roadmap

The roadmap is phased to align with course progression, with each phase building on the centralized data foundation established in the MVP.

Phase	Focus	Key Capabilities
Phase 1 — Foundation (MVP)	Core records & attendance	Student registration, course enrollment, attendance capture, basic reporting, role-based login
Phase 2 — Examinations	Assessment lifecycle	Exam scheduling, mark entry, grading rules, result publishing and approval workflow
Phase 3 — Reporting & Analytics	Institutional insight	Configurable reports, transcripts, performance dashboards, exportable data
Phase 4 — Extensions & Integration	Ecosystem connectivity	Notifications (email/SMS), institution ERP integration, mobile app enhancements

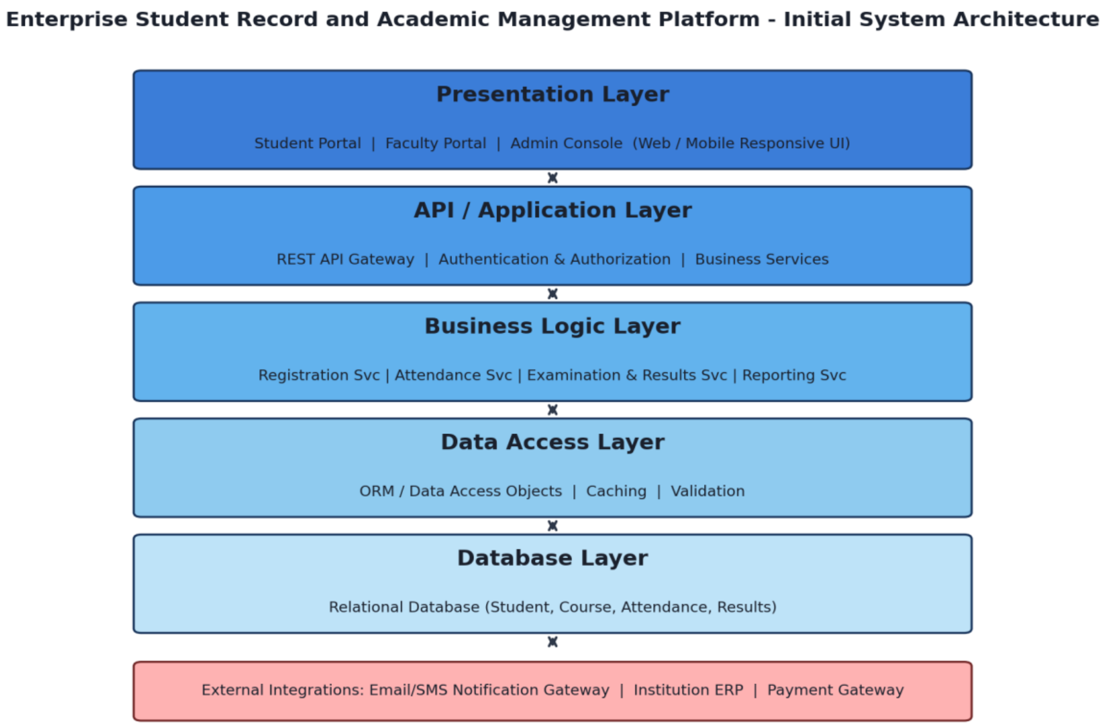
11.1 Alignment with Course Progression

Phase 1 corresponds to the planning and MVP definition completed in this unit; Phases 2–4 are intended to be built out incrementally as Agile sprints in subsequent units, using the backlog derived from the Functional Requirements Specification.

Deliverable 12

12. Initial System Architecture Diagram

The Platform follows a layered architecture to keep presentation, business logic, and data concerns independently maintainable and scalable.



12.1 Layer Descriptions

- Presentation Layer: Student portal, faculty portal, and admin console via a responsive web UI.
- API / Application Layer: REST API gateway handling authentication, authorization, and routing to business services.
- Business Logic Layer: Independent services for registration, attendance, examinations & results, and reporting.
- Data Access Layer: ORM/data-access objects, caching, and validation between services and the database.
- Database Layer: Relational database storing student, course, attendance, and result data.
- External Integrations: Notification gateway, institution ERP, and payment gateway, connected via the API layer in later roadmap phases.

Deliverable 13

13. Project Charter

Field	Details
Project Name	Enterprise Student Record and Academic Management Platform
Project Purpose	Replace manual academic administration with a centralized digital platform covering registration, attendance, examinations, and reporting.
Project Objectives	Centralize student data; reduce administrative turnaround time; improve data accuracy; enable self-service access; support data-driven decisions.
Scope Summary	MVP: registration, attendance, basic reporting. Later phases: examinations, analytics, integrations (see Product Roadmap).
Key Stakeholders	Students, Faculty Members, Academic Administrators, Examination Cell Staff, IT Support Personnel, Institution Management
High-Level Milestones	M1: Business understanding & Design Thinking complete. M2: Requirements & Product Vision finalized. M3: MVP defined & roadmap approved. M4: Repository initialized, ready for Agile development.
Success Criteria	Approved BRD, FRS, NFRS, and Product Vision; MVP scope agreed by stakeholders; repository and folder structure initialized.
Assumptions	Stakeholder availability for requirement validation; stable connectivity; incremental delivery across roadmap phases.
Risks	Scope creep beyond MVP; incomplete legacy data for migration; changing institutional policy requirements mid-project.

Deliverable 14

14. Project Repository Initialization

The following steps document the initialization of the version-controlled project repository.

14.1 Steps Performed

- Create the project root folder for the Enterprise Student Record and Academic Management Platform.
- Initialize a Git repository: git init
- Create a README.md describing the project purpose and structure.
- Create a .gitignore file appropriate to the technology stack (dependencies, build artifacts, environment files).
- Stage and commit the initial project structure: git add . followed by git commit -m "Initial commit: project structure and planning docs"
- Create and connect a remote repository, then push the initial commit: git remote add origin <repository-url> and git push -u origin main

14.2 Branching Convention (for subsequent Agile units)

- main — stable, release-ready code.
- develop — integration branch for ongoing work.
- feature/<name> — individual feature branches created from develop.

Deliverable 15

15. Project Folder Structure

The repository is organized to separate documentation, backend, frontend, and configuration concerns:

```
student-academic-platform/ |-- docs/ |   |-- BRD.md |   |-- product-vision.md |   |--
stakeholder-analysis.md |   |-- personas.md |   |-- FRS.md |   |-- NFRS.md |   |-- design-
thinking-report.md |   |-- roadmap.md |   `-- architecture-diagram.png |-- backend/ |   |--
src/ |   |   |-- modules/ |   |   |   |-- registration/ |   |   |   |-- attendance/ |   |   |
|-- examinations/ |   |   |   `-- reporting/ |   |   `-- common/ |   `-- tests/ |-- frontend/
|   |-- src/ |   |   |-- pages/ |   |   |   |-- student-portal/ |   |   |   |-- faculty-
portal/ |   |   |   `-- admin-console/ |   |   `-- components/ |   `-- public/ |-- config/ |-
- .gitignore `-- README.md
```

15.1 Rationale

- docs/ holds all planning artifacts produced in this unit (BRD, FRS, NFRS, roadmap, diagrams).
- backend/ and frontend/ are pre-structured by module (registration, attendance, examinations, reporting) to mirror the Functional Requirements Specification.
- Separation of concerns supports independent development and testing of each module in future Agile sprints.

Deliverable 16

16. Product Discovery Documentation

16.1 Discovery Methods

- Process study: reviewing the existing manual academic-management workflow end to end (admissions through result publishing).
- Stakeholder interviews: conversations with representative students, faculty, and administrative staff.
- Problem framing: consolidating pain points into a single problem statement (see Design Thinking Report, Section 7.2).

16.2 Key Findings

- Data about the same student is re-entered in multiple departments, creating inconsistency risk.
- Attendance tracking is the most frequent daily task for faculty and the most time-sensitive for students.
- Result publishing delays are a recurring source of frustration across all stakeholder groups.
- Administrators lack a consolidated, real-time view of institution-wide academic metrics.

16.3 Validated Problem Statement

Students, faculty, and administrators at the institution lack a centralized, reliable system to manage academic records, leading to delays, data inconsistency, and administrative overhead across registration, attendance, and examination processes.

16.4 Discovery Outcome

These findings directly informed the Business Requirements Document, the Stakeholder Analysis, the User Personas, and the MVP scope, ensuring the planned platform addresses the highest-impact problems first (registration and attendance) before extending to examinations and analytics.